MIT ENERGY & CLIMATE HACK 2026 · REON CHALLENGE

PeakSure

Know what your battery can deliver when it counts

Cell-level intelligence with REON EBS for the batteries Massachusetts towns use at peak hours: prove it on today's batteries, then build it into tomorrow's.

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Independent analysis, not endorsed by REON

THE PROBLEM

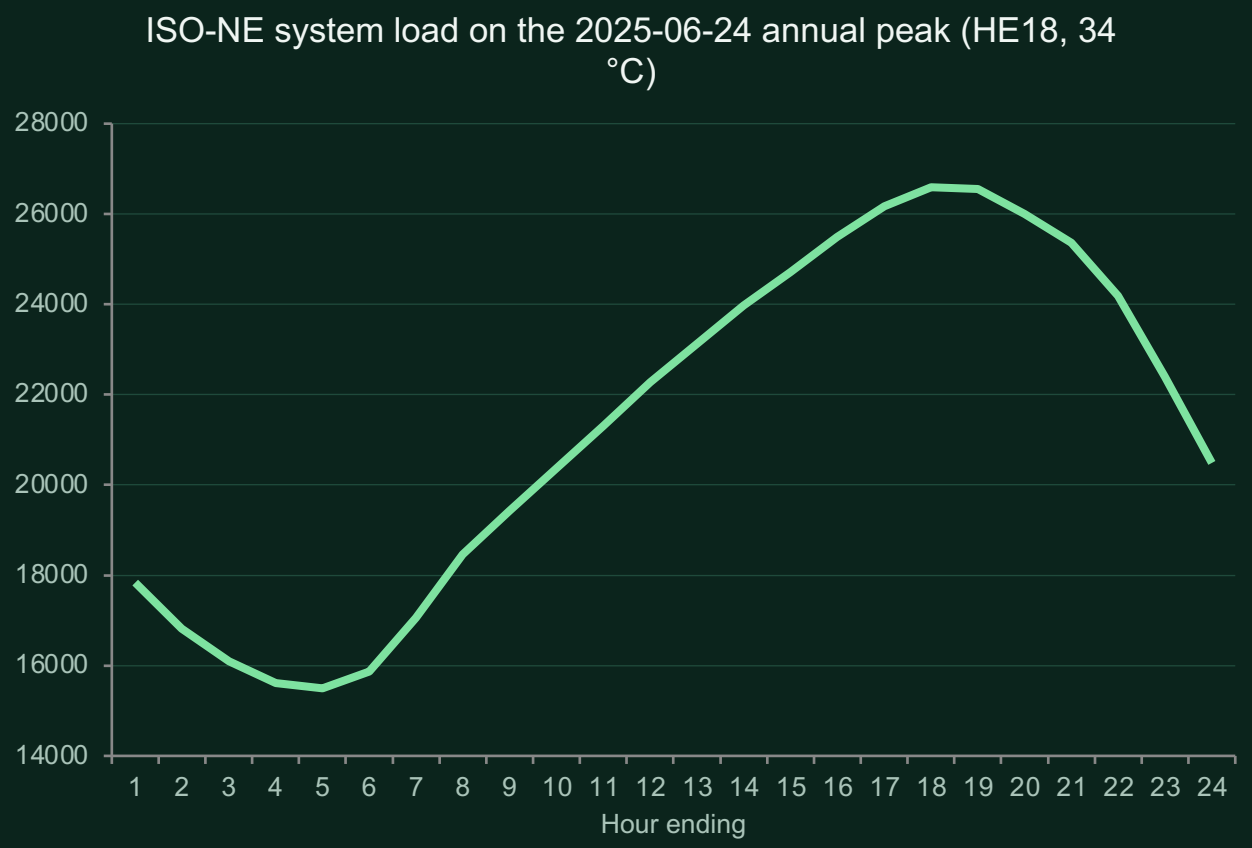
One summer hour decides a town's power bill

Massachusetts towns pay ISO-NE transmission ($183.71/kW-yr) on 12 monthly peaks and capacity ($3.58/kW-month) on the annual peak. Those charges reach every household.

Today, operators guess. Promise too much and the battery runs dry before the peak ends; too little and savings stay on the table.

$244,460

saved by Sterling's battery at one annual peak (Community Resilience Center, 2017)



Our tool and video use this real 2025 peak (data: ISO-NE 2019–2025). The 2026 peak was 2 July, HE19, 25,361 MW (ISO-NE, Aug 2026).

WHERE REON COMES IN

One weak cell stops a whole string. REON switches it out

Conventional string: stops when the 70 Ah cell is empty



REON bypass: weak cell switched out, the rest keep going



+20% energy on the challenge's 16-cell example (84.06 vs 70 Ah per cell)

DRAIN TEST · same cells, full power until every string stops

638 kWh

stranded by fixed strings

204 kWh

stranded with REON bypass

Delivered: 3,855 kWh conventional vs 4,212 kWh REON EBS

Modelled plant: 8 strings × 240 cells (simulation abstraction, not REON's design), battery age 9 years. Simulated.

OUR SOLUTION · PHASE 1 · RETROFIT

1 Retrofit → 2 Scale

Add cell-level control to batteries that already exist

COST

$25–75

per kWh (team estimate). No new battery to buy.

Exit test: more energy delivered and fewer missed peaks, measured against the same battery's history.

WHERE IT PAYS · extra savings per year (model)

Faulty cells (0.2%/yr)	**+$154K**
Failing cooling	**+$98K**
1-hour battery	**+$43K**
4-hour battery	**+$34K**
Healthy 2-hour battery	**−$0.6K**

MASSACHUSETTS CANDIDATES

Sterling 2 MW / 3.9 MWh (2016)

Reading 5 MW / 10 MWh (2019)

Holden 5 MW / 22 MWh (2024)

With proof in hand, sell REON EBS with cell control

When

- Phase 1 shows measured extra energy and fewer missed peaks
- REON confirms EBS specifications and certification

Where

- New municipal light plant storage in Massachusetts
- Commercial sites on demand charges, e.g. National Grid G-2 at $15.24/kW-month

How

- Priced -5% to +20% vs a conventional battery (team estimate)
- PeakSure ships as the decision layer

Economics: team sheet 3 MW / 9 MWh (167 EBS): upfront ≈ $4.0M, NPV +$0.91M at $500K/yr savings · our simulation, new 2.5 MW / 5 MWh plant: NPV $1.20M conventional, $0.88M EBS + PeakSure at assumed prices

HOW THE MODEL WORKS

Real peaks in, a dollar answer out

1 Real data

ISO-NE hourly peaks and temperatures, 2019–2025

2 Cell physics

Every cell ages, heats and drains over 15 years

3 Digital twin

Estimates what each cell can still deliver

4 Commit and score

95% commitment, scored on kW actually delivered

Three cases on identical cells

A · Conventional: fixed strings, one pack test a year

B · REON EBS: cell bypass, same operator rule

C · REON EBS + PeakSure: twin of every cell, commits with 95% confidence

CONFIRMED	ISO-NE peaks and rates; EBS 30 kW / 60 kWh
DERIVED	2 h = 5 MWh ÷ 2.5 MW, nominal only
LITERATURE	Na-ion ageing (TUM lab data)
ASSUMED	8 × 240 cells, switch loss, DC floor

Monte Carlo: 64 paired runs per case on a modelled 2.5 MW / 5 MWh plant (about 84 EBS blocks). The browser model matches the Python model to 0.001 kWh.

RESULTS

It pays where cells limit delivery

≈ $0

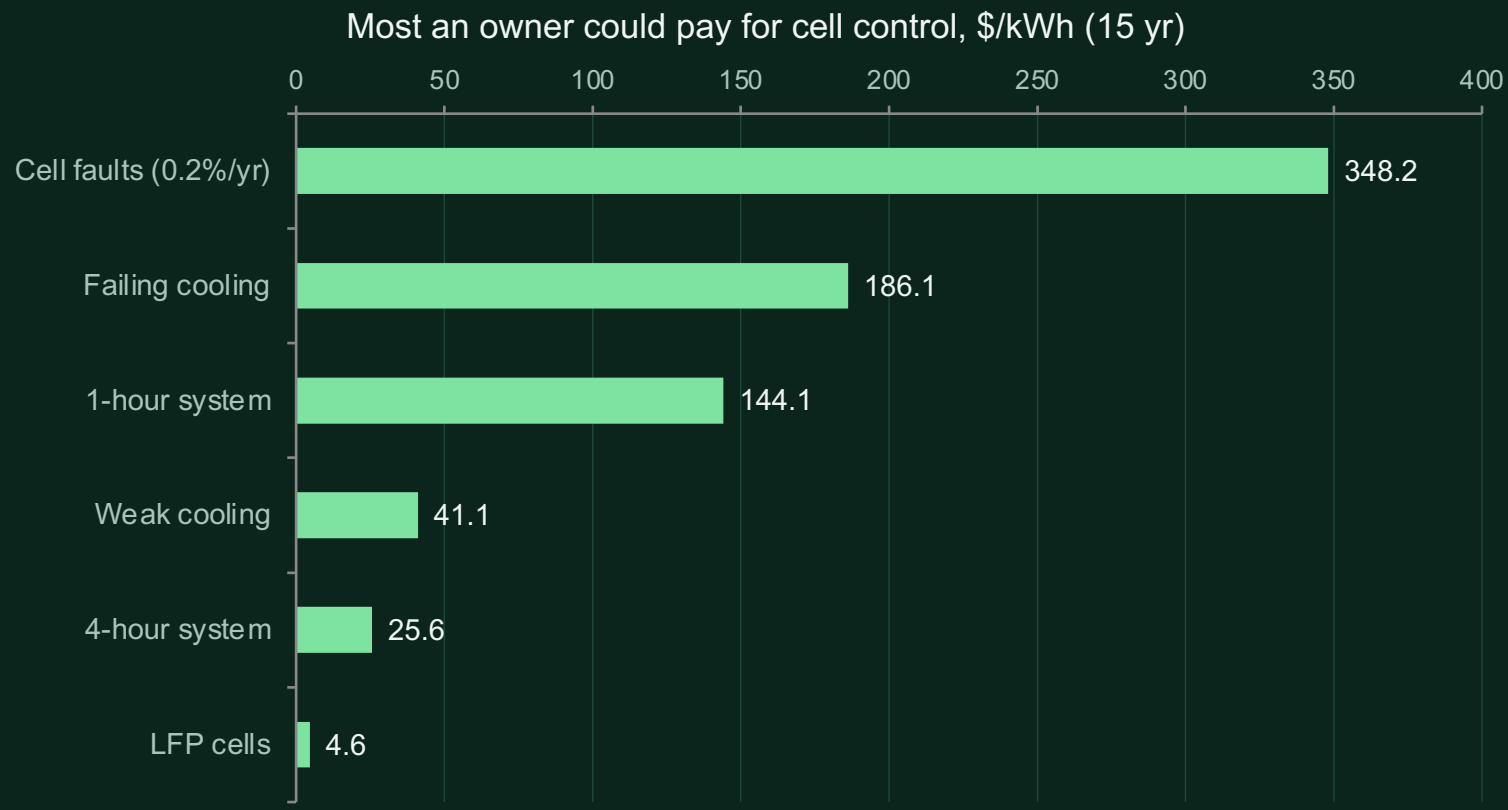
2-hour plant, normal duty: bypass + twin
−$566/yr. Peak timing matters more.

+$33.8K/yr

4-hour system (hypothetical): bypass + twin

75% vs 66%

monthly peaks covered: PeakSure's 95% commitment vs conventional



Where it does not pay: Switch loss 2% -5.6 · High ageing spread -16.6 · Many dispatch days -19.0 $/kWh

BATTERY OPTIONS

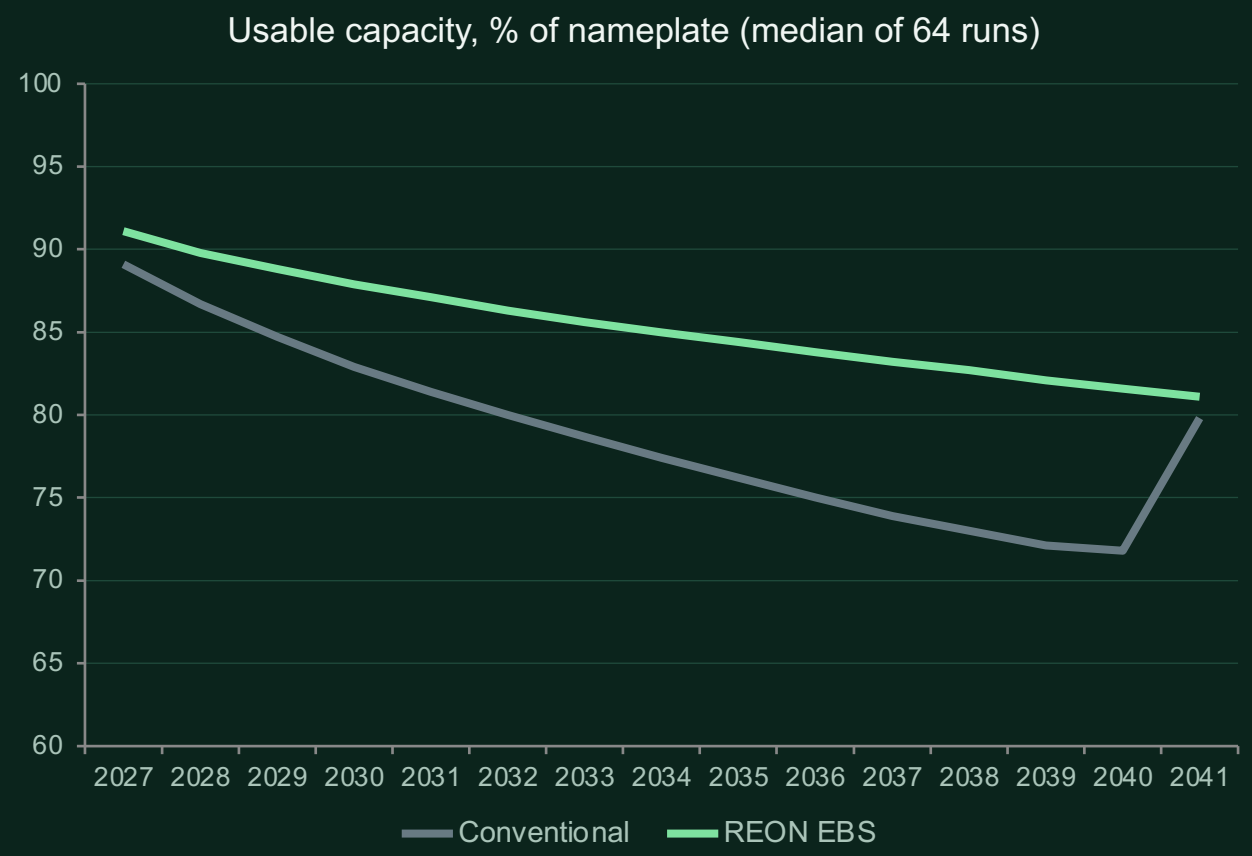
One control system, two cell types

	Lithium-ion · REON EBS Li+ (modelled as LFP)	Sodium-ion · REON EBS Na+
Energy per space	High: REON says “more than twice the energy in the same form factor”	Lower
Cold weather	Weaker	Better: REON rates −40 °F
Installed price (team estimate)	$219–846/kWh	$251–969/kWh
Cell-control premium (our model)	$4.6/kWh	$0.3/kWh (base case)
Materials	Lithium, iron; 56 kg CO_2e/kWh to make	Abundant sodium, no lithium
Best fit	Town storage, tight space	Cold or indoor sites, high cycling

REON's BMS works with partner cells · Sources: REON; Tarascon (2020); Vaalma et al. (2018); Llamas-Orozco et al. (2023)

IMPACT AND SUSTAINABILITY

Same cells, more energy, for longer



70% = service threshold: cells are topped up below it

FOR THE TOWN

$20.93

saved per Sterling customer at one hot-day peak ($84.0K total, live tool)

FOR RELIABILITY

4,212 kWh

from the same cells with bypass, vs 3,855 kWh: more backup hours

FOR THE PLANET

≈ 7.0 t CO_2e

141 kWh of top-up cells avoided over 15 years (4.4–12.9 t); capacity stays ≥ 81% vs 72%

OUR RECOMMENDATION

Start with a retrofit pilot on struggling batteries

NEXT 3 MONTHS

Read-only data pilot

Replay one REON battery log; PeakSure recommends each peak commitment, people decide.

3–12 MONTHS

Retrofit pilot

REON cell-level BMS on 1–2 Massachusetts batteries; measure delivered energy, missed peaks and savings.

AFTER THE EVIDENCE

Scale with REON EBS

New EBS systems with cell control and PeakSure built in, priced on measured value.

Compliance in Massachusetts

- Siting: 225 CMR 29.00 and local permits
- Fire: 527 CMR 1.00, NFPA 855, UL 9540 / 9540A
- Electrical: 527 CMR 12.00; interconnection with the host utility
- End of life: 310 CMR 30.000

OUR ASK TO REON

One anonymized battery log, cells per block, DC floor and switch-loss range: that turns this model into evidence.